



Lucas Lemos Ricaldoni

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● ABOUT ME

Computer Engineering student at Politecnico di Milano with a technical background in Industrial Automation. I possess a strong portfolio in robotics, IoT, embedded programming (C/C++, Python), and 3D modeling (Fusion 360). Experienced in integrating hardware and software to solve real-world problems. Recognized for technical excellence, including the 1st Place at the CI-IA Health Challenge (Unimed BH) for developing the engineering and prototyping of an AI-applied hospital screening solution, and awards in autonomous robotics (CoRA, OBR). I am highly motivated to apply my practical expertise in embedded systems, computer vision, and rapid prototyping to innovative AI and technology projects.

● EDUCATION & TRAINING

14/09/2026 - Current - CREMONA, ITALY

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING- POLITECNICO DI MILANO

Level in EQF: 6

31/12/2022 - 31/12/2025 - BELO HORIZONTE, BRAZIL

HIGH SCHOOL DIPLOMA WITH PROFESSIONAL QUALIFICATION IN INDUSTRIAL AUTOMATION- TECHNICAL COLLEGE OF UFMG

Technical training focused on **Industrial Automation**, **Robotics**, and **Embedded Systems**. **Electronics and Programming:** Developing circuits and programming in C/C++ and Python for microcontrollers like Arduino and ESP32. **Robotics:** Designing and programming autonomous line-follower robots using PID control. **3D Modeling:** Expert use of SolidWorks and Fusion 360 for mechanical design and prototyping. **Industrial Control:** Knowledge of PLC programming (Ladder, ST, SFC) and SCADA systems

Level in EQF: 4

● WORK EXPERIENCE

28/02/2025 - 31/01/2026 - BELO HORIZONTE, BRAZIL

JUNIOR RESEARCHER & PROTOTYPING INTERN LIPS - INNOVATION, PROTOTYPING AND SIMULATION LABORATORY (UFMG)

Engineered a vibrating smart watch prototype with an integrated moisture sensor (IoT) tailored for elderly care, awarded "Academic Relevance" at the XV UFMG Scientific Initiation Seminar.

Created functional MVPs and final products by successfully integrating electronics, automation logic, and mechanical design.

Developed the electrical panel, power distribution infrastructure, and control logic (using Autostation Pro and IHM Tools) for an automated industrial machine as part of a partnership project with Petrobras.

01/01/2026 - CURRENT

FOUNDER & IOT/PROTOTYPING DEVELOPER VITRUR

Developed integrated Internet of Things (IoT) and Robotics solutions, combining 3D modeling, hardware development, and programming logic (e.g., engineered an automated smart IoT device).

Executed advanced CAD modeling using Fusion 360 and SolidWorks, from structural electronic enclosures to rapid prototyping and final functional parts.

Operated advanced 3D printing equipment using technical, high-performance filaments (such as Tritan HT) to ensure mechanical and thermal compliance for demanding environments.

Manufactured and delivered over 100 technical and functional parts, meeting custom engineering demands with a focus on high durability and dimensional accuracy.

● **SKILLS**

Programming Languages

Python (computer programming) | C | C++ | MATLAB | JavaScript

Engineering & Modeling

robotics | Fusion 360 | Matlab Environment (Matlab, Simulink, Simscape) | SolidWorks | Creality Print | use CAD software | printing techniques | 3D printing process

Development Tools

Microsoft Excel | LaTeX | Git & GitHub | Arduino IDE & Visual Studio Code

● **LANGUAGE SKILLS**

Mother tongue(s): **PORTUGUESE**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ITALIAN	B2	B2	B2	B2	B2
ENGLISH	B2	B2	B1	B1	B1

● **HONOURS AND AWARDS**

- 26/09/2025 Center for Artificial Intelligence in Health (CI-IA)
1st Place - Geraldo Ribeiro Award | CI-IA Health Challenge
[Developed the winning 3D modeling and mechanical simulations for an innovative hospital screening technological solution.](#)
- 18/10/2025 Federal University of Minas Gerais (UFMG)
Academic Relevance Award - UFMG Knowledge Week 2025
[Recognized for outstanding academic contribution and research quality during the annual UFMG scientific showcase. Developed a complete IoT system designed for elderly care, featuring a smart device that monitors plant humidity and sends automated alerts to ensure a healthy environment for the user](#)
- 18/09/2025 Federal University of Minas Gerais (UFMG)
3rd Place (Advanced Line Follower) - CoRA 2025
[As Captain of the "Tartaruga Linhas" team, led the development of autonomous robots using Arduino and C/C++. Won 3rd place in the competitive advanced line-follower category.](#)
- 18/09/2025 Federal University of Minas Gerais (UFMG)
1st Place - Best Design CoRA 2025
[As Captain of the "Tartaruga Linhas" team, led the development of autonomous robots using Arduino and C/C++. Won 1st place for technical design.](#)

● CERTIFICATIONS

31/12/2025 Harvard University

Applications of TinyML

Online - Professional certification covering the deployment of machine learning models on low-power, resource-constrained microcontrollers (24 hours)

30/11/2024 Federal Institute of Minas Gerais (IFMG)

Industry 4.0: Internet of Things

Online -

[Focused on Industrial Internet of Things \(IIoT\) and general IoT concepts, covering sensors, connectivity, and dataapplication in industrial environments \(30 hours\).](#)

30/11/2024 OpenCV University

OpenCV Bootcamp

Project Based -

[Practical certification focused on image processing, object detection, and computer vision algorithms \(10 hours\).](#)

30/11/2023 Zero to Mastery (Udemy)

Complete A.I. & Machine Learning, Data Science Bootcamp

Online -

[Complete bootcamp covering the entire Data Science workflow using Python. Acquired practical skills in MachineLearning \(Supervised/Unsupervised Learning\), Deep Learning with TensorFlow and Keras, and data analysis usingPandas, NumPy, and Matplotlib. \(44 hours\)](#)

30/11/2024 Harvard University

CS50's Web Programming with Python and JavaScript

Project Based -

[> Comprehensive course on designing and implementing web apps with Python, JavaScript, and SQL using frameworkslike Django and React \(80 hours\).](#)

31/12/2024 IBM (International Business Machines Corporation)

Introduction to Computer Vision and Image Processing

Online - Technical course focused on the fundamentals of Computer Vision. Covered image processing techniques, object detection, and image classification algorithms using Python and OpenCV. (22 hours)

30/11/2023 Harvard University

CS50's Introduction to Programming with Python

Online -

[Comprehensive introduction to programming with Python, covering functions, variables, conditionals, loops, andobject-oriented programming \(OOP\). Developed hands-on projects focusing on code readability, debugging, andtesting unit libraries. \(40 hours\)](#)

● CONFERENCES & SEMINARS

13/10/2025 - 17/10/2025 Federal University of Minas Gerais (UFMG)

UFMG Knowledge Week 2025 Presented a scientific initiation project focused on Internet of Things (IoT) solutions for elderly care. Demonstrated a functional prototype capable of monitoring environmental data and sending automated alerts, developed during my internship at the Innovation, Prototyping, and Simulation Laboratory (LIPS) UFMG.